

Probeklausur Beispiele

1.3 $Q = 400 - P^{-2}$ $TC = 0.625 \cdot Q^2$ $MR = MC$
 $Q = \frac{400}{P^2}$ $MC = 1.25Q$ $\frac{10}{\sqrt{Q}} = 1.25 \cdot Q$
 $P^2 = \frac{400}{Q}$ $R = (20 \cdot Q^{-1/2}) \cdot Q$ $10 = 1.25Q \cdot \sqrt{Q}$
 $P = \sqrt{\frac{400}{Q}}$ $= 20Q \cdot Q^{1/2}$ $8 = Q \cdot \sqrt{Q}$
 $P = \frac{20}{\sqrt{Q}}$ $MR = 20 \cdot \frac{1}{2} Q^{-1/2}$ $8 = Q^{3/2}$
 $P = \frac{20}{Q^{1/2}}$ $= 10 \cdot Q^{-1/2}$ $Q = \sqrt[3]{8} = 2$
 $P = 20 \cdot Q^{1/2}$ $= \frac{10}{\sqrt{Q}}$

1.5 Produzentenrente: $\frac{60 \cdot (50 - 20)}{2} = 900$

Quiz Fragen

1 a) $s_1 s_2 = (\text{oben, links})$
 $s_1 s_2 = (\text{unten, rechts})$

b) P1: $s_1 s_2 = (\text{mitte, mitte})$
 Kooperation: $s_1 s_2 = (\text{unten, links})$ $7+4=11$
 Abweichung: $s_1 s_2 = (\text{oben, rechts})$ $8+1=9$

c) $\delta < 1$

KOOP: $7 + \frac{4}{1-\delta}$

Abw: $8 + 1 \cdot \frac{\delta}{1-\delta}$

$7 + \frac{4}{1-\delta} > 8 + \frac{\delta}{1-\delta}$

$7 + 4 \cdot \frac{1}{1-\delta} > 8 + \frac{\delta}{1-\delta}$ $| -7$

$4 \cdot \frac{1}{1-\delta} > 1 + \frac{\delta}{1-\delta}$ $| \cdot (1-\delta)$

$4 > 1 + \delta$

$\delta > 3$

2 $Q(P) = 600 - 3P$, $MC = 20$, $P = \frac{600-Q}{3}$

b) $\pi = (200 - \frac{1}{3}Q) \cdot Q - 20 \cdot Q$

$\pi' = 200 - \frac{2}{3}Q - 20 = 0$

$180 = \frac{2}{3}Q$

$Q = 270$

$q_1 = 135$

$q_2 = 135$

$\pi = 24300$

$\pi_1 = 12150$

$\pi_2 = 12150$

d) $\frac{12150}{1-\delta} > 24300$

$12150 > 24300 \cdot (1-\delta)$

$\delta > 0.5$

c) $\pi (110 - \varepsilon - 20) \cdot Q = 80 \cdot 270 = 21600$

$$\textcircled{1} \quad Q = 250 - 0.5P, \quad TC = 50Q + 5.5Q^2$$

$$\pi = (500 - 2Q) \cdot Q - 50Q - 5.5Q^2$$

$$\pi' = 500 - 4Q - 50 - 11Q = 0$$

$$450 = 15Q$$

$$Q = 30$$

$$\pi = 6750$$

$$\textcircled{2} \quad E(ux) = 0.6 \cdot 20 + 0.4 \cdot 220 = 100$$

$$u(100) = 100$$

$$\textcircled{3} \quad Q = 400P^{-2}$$

$$Q = \frac{400}{P^2}$$

$$P^2 = \frac{400}{Q}$$

$$P = \frac{20}{\sqrt{Q}}$$

$$P = \frac{20}{Q^{\frac{1}{2}}}$$

$$P = 20 \cdot Q^{-\frac{1}{2}}$$

$$R = P \cdot Q$$

$$= (20 \cdot Q^{-\frac{1}{2}}) \cdot Q$$

$$= 20Q \cdot Q^{\frac{1}{2}}$$

$$MR = 20 \cdot \frac{1}{2} Q^{-\frac{1}{2}}$$

$$= 10 \cdot Q^{-\frac{1}{2}}$$

$$MR = MC$$

$$10 \cdot Q^{-\frac{1}{2}} = 1.25Q$$

$$\frac{10}{\sqrt{Q}} = 1.25Q$$

$$10 = 1.25Q \cdot \sqrt{Q}$$

$$10 = 1.25Q^{\frac{3}{2}}$$

$$8 = Q^{\frac{3}{2}}$$

$$\sqrt[3]{8} = Q$$

$$Q = 4$$

$$\textcircled{5} \quad \max p: 50$$

$$\max Q = 60$$

$$MC = 20$$

$$\frac{60 \cdot (50 - 20)}{2} = 900$$

$$\textcircled{6} \quad U(w) = 10w^{0.5}$$

$$E(ux) = 0.4 \cdot 10 \cdot 25^{0.5} + 0.6 \cdot 10 \cdot 100^{0.5} = 80$$

$$80 = 10 \cdot w^{0.5}$$

$$8 = \sqrt{w}$$

$$w = 64$$

$$\textcircled{7} \quad P = 20 - 2Q, \quad C = 4q_i$$

$$\pi = (20 - 2(q_1 + q_2)) \cdot q_1 - 4q_1$$

$$\pi' = 20 - 4q_1 - 2q_2 - 4 = 0$$

$$16 = 4q_1 + 2q_2$$

$$\frac{16 - 2q_2}{4} = q_1 = q_2 = q$$

$$q = \frac{16 - 2q}{4}$$

$$4q = 16 - 2q$$

$$6q = 16$$

$$q = 2.667 = q_1 = q_2$$

$$\pi_1 = 14.2224 = \pi_2$$

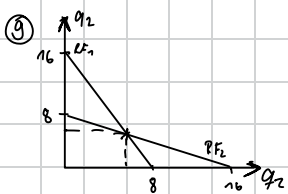
$$\textcircled{8} \quad \pi = (20 - 2Q) \cdot Q - 4Q$$

$$\pi' = 20 - 4Q - 4 = 0$$

$$16 = 4Q$$

$$Q = 4 \quad q_1 = 2; q_2 = 2$$

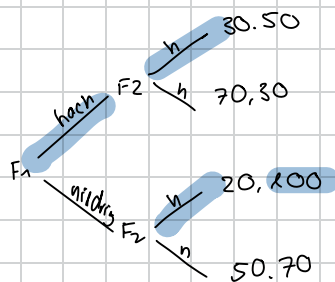
$$\pi = 32 \quad \pi_1 = 16, \pi_2 = 16$$



⑩ a) $s_1 s_2 = (\text{hoch}, \text{hoch})$

- > Kooperation würde zu besserem Ergebnis führen
- > Gefangenendilemma

b)



$F_1: s_1 = (\text{hoch}, \text{niedrig})$

$F_2: s_2 = (hh, hn, nh, nn)$

SPNE: $s_1 s_2 = (\text{hoch}, \text{hoch})$

⑪ $\pi_w = 44 \cdot r - 2r^2$

$\pi_g = 64 - 2g^2 - 2rg$

1 Unternehmen: $\pi = \pi_w + \pi_g$

$\pi = 44r - 2r^2 + (64g - 2g^2 - 2rg)$

$\pi'_r = 44 - 4r - 2g$

$\pi'_g = 64 - 4g - 2r$

$44 - 4r - 2g = 0$
 $\frac{44 - 2g}{4} = r$

$64 - 4g - 2r = 0$
 $\frac{64 - 2r}{4} = g$

$\frac{44 - 2 \cdot \frac{64 - 2r}{4}}{4} = r$

$44 - 2 \cdot \frac{64 - 2r}{4} = 4r$
 $44 - 32 + r = 4r$

$12 = 3r$

$r = 4$

$g = 14$

⑫ $\pi_w = 44r - 2r^2 - 2rg$

$\pi_g = 64g - 2g^2 - 2rg + 2rg$

$\pi'_g = 64 - 4g = 0$

$64 = 4g$

$g = 16$

$\pi'_w = 44 - 4r - 2g = 0$

$\frac{44 - 2g}{4} = r$

$r = 3$

Quiz 2

$Q = 160 - 2P$, $C = 1200 + 8 \cdot Q$

$P = \frac{160 - Q}{2}$

a) $\pi_1 = (80 - \frac{1}{2}(q_1 + q_2)) \cdot q_1 - 8 \cdot q_1 - 1200$

$\pi'_1 = 80 - q_1 - 0.5q_2 - 8 = 0$

$72 = q_1 + 0.5q_2$

$72 - 0.5q_2 = q_1$

$q_2 = 72 - 0.5q_1$

$\pi_1 = -48$

$\pi_2 = -48$

$q_2 = 72 - 0.5 \cdot (72 - 0.5q_2)$

$q_2 = 72 - 36 + \frac{1}{4}q_2$

$\frac{3}{4}q_2 = 36$

$q_2 = 48$

$q_1 = 48$

↳ Nein Beide Firmen machen Verluste